To What Extent Do a Song's Melody and Lyrics Elicit Similar Emotional Responses in Listeners?

A Literature Review

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Abstract

This literature review surveys existing research relevant to the question of whether a song's melody and lyrics elicit similar emotional responses in listeners. We synthesise findings from lyric-based music classification, topic modelling for lyrics, symbolic music representation through ABC notation, and music emotion recognition from both audio and textual modalities. The review further catalogues publicly available datasets that provide lyrics, melodic notation, metadata, and emotion labels, assessing their suitability for a combined lyric–melody emotion analysis. Across the surveyed studies, a recurring finding is that lyrics and melodies each carry independent emotional signals, and that combining textual and audio features consistently yields the strongest classification performance. We identify methodological gaps in existing work—particularly the scarcity of studies that directly align symbolic melodic representations with lyric-level emotion features—and position our planned research within this landscape.

1 Introduction

Music communicates emotion through two primary channels: melody and lyrics. While listeners intuitively perceive both components, the extent to which each modality contributes to the overall emotional experience remains an active area of inquiry in text analytics, music information retrieval (MIR), and affective computing. The central research question guiding this review is:

> *To what extent do a song's melody and lyrics elicit similar emotional responses in listeners?*

Answering this question requires drawing on several distinct but intersecting strands of research. First, it demands an understanding of how lyrics can be computationally represented and classified—work that spans natural language processing (NLP) and text analytics. Second, it requires knowledge of symbolic music representations, particularly ABC notation, which encodes melody in a machine-readable text format amenable to computational analysis. Third, it necessitates engagement with the music emotion recognition (MER) literature, which has explored the relative contributions of audio and textual features to emotion classification. Finally, practical investigation depends on the availability of suitable datasets that pair lyrics with melodic information and emotion labels.

This review is structured as follows. Section 2 examines lyric-based music analysis and classification. Section 3 discusses topic modelling approaches applied to lyrics. Section 4 reviews ABC notation and symbolic music representation. Section 5 surveys research on the emotional roles of lyrics and melodies. Section 6 considers audio and melodic feature extraction for emotion recognition. Section 7 catalogues available datasets and data sources. Section 8 synthesises the findings and identifies research gaps, and Section 9 concludes the review.

2 Lyric-Based Music Analysis and Classification

A foundational study for lyric-based music analysis is the work of Fell and Sporleder (2014), who investigated whether song lyrics alone can classify music by genre, perceived quality, and publication era. Using a large corpus of approximately 400,000 English-language lyrics drawn from 7,200 artists on Lyricsmode.com, the authors engineered 13 feature classes organised across five linguistic dimensions: *vocabulary* (type–token ratio, slang from Urban Dictionary, rare words), *style* (POS and chunk tags, rhyme patterns, echoisms, line lengths), *semantics* (categories from the Regressive Imagery Dictionary covering emotional, primordial, and conceptual thought), *orientation* (pronoun ratios,

tense distributions), and *song structure* (chorus detection, repetitive blocks, title repetition).

Using Support Vector Machine (SVM) classifiers with artist-disjoint train/test splits, Fell and Sporleder (2014) achieved the following results across three classification tasks:

- **Genre classification** (8 genres): A combined feature set (n-grams plus extended features) achieved 52.5% F1, compared to a 49.0% n-gram baseline. Rap was the most distinguishable genre (77.6% F1), while Folk proved the most difficult (29.6% F1).
- **Quality prediction** (best vs. worst songs for Pop/Rock, Metal, and Rap): Combined features reached 75–86% F1, with higher-quality songs exhibiting greater vocabulary diversity and fewer first-person pronouns.
- **Era classification** (Pop/Rock songs from three periods): Combined features achieved 46.8% F1. Newer songs tended to have longer lyrics and fewer choruses.

These results demonstrate that lyrics carry strong genre and quality signals through function words, POS patterns, and structural repetition—features that are directly relevant to our research, as they show that textual content encodes musically meaningful information independently of the audio signal.

3 Topic Modelling and Retrieval in Lyrics

Complementing classification approaches, Sasaki et al. (2014) developed LyricsRadar, a lyrics retrieval system that applies Latent Dirichlet Allocation (LDA) to model the latent topic structure of song lyrics. Each song's lyrics are represented as a five-dimensional topic vector, enabling both topic-based search and inter-song similarity computation.

The system employs t-SNE (van der Maaten and Hinton, 2008) to project the high-dimensional topic vectors onto a two-dimensional plane, positioning songs with similar topic distributions closer together. Visual encoding uses RGB colour mapping compressed from the five topic dimensions, such that songs with similar thematic profiles appear similar in both position and colour. An interactive radar chart displays the topic mixture for each song as a pentagon, with each vertex corresponding to one of the five learned topics.

LyricsRadar is relevant to our investigation for several reasons. Its LDA-based representation captures latent semantic content in lyrics that goes beyond surface-level word frequencies, providing a principled means of quantifying lyrical themes. The dimensionality reduction and similarity measurement framework could be adapted to compare lyric-derived topic profiles with emotion ratings derived from melodic analysis, enabling a cross-modal comparison at the song level.

4 ABC Notation and Symbolic Music Representation

To analyse melody computationally, a machine-readable representation is required. ABC notation is a text-based shorthand for musical notation that encodes pitch, duration, key, metre, and structural markers using ASCII characters (Walshaw, 2014). A typical ABC transcription includes header fields—`X` (index), `T` (title), `M` (time signature), `L` (default note length), `K` (key)—followed by the melodic body, where letters `A`–`G` (uppercase) and `a`–`g` (lowercase, one octave higher) denote pitches, digits indicate duration multiples, and symbols such as `:|` and `|:` encode repeats.

Walshaw (2014) conducted a statistical analysis of the ABC corpus available through the abcnotation.com search engine, which at the time of the study comprised approximately 167,000 unique tunes. Of these, 12,574 (7.5%) included lyrics within the ABC transcription. The study analysed key signature distributions and introduced a method for melodic similarity detection based on pitch contour vectors constructed from quantised incipits (the first four bars), where each vector element stores the interval in semitones between the corresponding note and the first note of the melody.

This work establishes ABC notation as a viable modality for computational melody analysis and identifies available corpora. For our research, the pitch contour representation offers a quantifiable melodic feature that can be compared with emotion features derived from lyrics, providing a pathway to test whether melodic and lyrical emotions align.

5 Emotional Roles of Lyrics and Melodies

The most directly relevant strand of prior work examines whether lyrics and melodies function as equal emotional partners. Ali and Peynircioğlu (2006) investigated this question through four experiments designed to isolate the emotional contributions of each modality.

The researchers curated 32 instrumental excerpts (approximately 20 seconds each) from classical, jazz, and soundtrack recordings, categorised into four emotion quadrants based on the circumplex model of affect: *happy* (high arousal, positive valence), *sad* (low arousal, negative valence), *calm* (low arousal, positive valence), and *angry* (high arousal, negative valence). Sixteen sets of adapted lyrics were recorded by male and female vocalists and paired with the melodies under both congruent and incongruent conditions.

Key experimental designs included:

- **Experiment 1 (Congruency)**: A within-subjects design comparing emotional in-

tensity ratings for melodies alone versus melodies paired with emotionally congruent lyrics.

- **Experiment 2 (Mismatched stimuli)**: Incongruent pairings (e.g., happy melody with sad lyrics) to determine which modality dominates perceived emotion.
- **Experiments 3 and 4 (Emotion transference)**: Cross-modal association tasks where participants viewed neutral images while listening to music, to measure whether musical emotion transfers to unrelated objects.

Participants rated stimuli on nine-point scales for all four emotions, and the researchers employed ANOVA with post-hoc comparisons (t-tests, Scheffé tests) to analyse main effects and interactions. Their findings indicated that while both melodies and lyrics convey emotion, the melody tends to be the dominant carrier of emotional information, with lyrics modulating rather than overriding the melodic emotion. This asymmetry is a central motivation for our research question: if melody typically dominates, understanding the conditions under which lyrics exert stronger influence could reveal important aspects of how multimodal emotional processing operates in music.

6 Audio and Melodic Features for Emotion Recognition

The music emotion recognition (MER) literature has explored a variety of audio feature types. Rocha et al. (2013) distinguished between Standard Audio (SA) features and Melodic Audio (MA) features:

- **Standard Audio features** are low-level descriptors (LLDs) computed from short-time spectra of the audio waveform, typically extracted using toolkits such as MIR Toolbox and PsySound.
- **Melodic Audio features** require a prior melody transcription step and encompass three categories: pitch and duration, vibrato characteristics, and contour typology.

Using data from the AllMusic database and classification algorithms including SVM (SMO and LibSVM), K-Nearest Neighbours, C4.5, Bayes Network, Naïve Bayes, and Simple Logistic, Rocha et al. (2013) found that melodic features outperformed standard audio features, and that combining both types yielded the best classification accuracy. This result is significant for our study, as it suggests that melodic contour—the feature most analogous to what ABC notation encodes—carries particularly rich emotional information.

In a complementary study, Malheiro et al. (2013) specifically investigated emotion recognition from lyrics using a dataset of 764 samples with paired audio and lyric data. Lyrics were tagged with emotion labels sourced from AllMusic, and lyric text was retrieved from lyrics.com, ChartLyrics, MaxiLyrics, and via the Google API. Features were derived from established NLP frameworks including Jlyrics, Synesketch, and ConceptNet, alongside bag-of-words representations. Their best lyric-only model achieved 44.2% F-measure using SVMs, but a bi-modal analysis combining audio and lyric feature sets consistently outperformed either modality alone. This finding reinforces a pattern observed across the MER literature: lyrics and audio provide complementary emotional signals, and their fusion improves recognition accuracy.

7 Available Datasets and Data Sources

[1] How about the title "Data Sources Evaluation and Data Alignment"

A critical practical consideration for our research is the availability of datasets that provide lyrics, melodic representations, metadata, and emotion labels. Table 1 summarises the datasets identified during our survey.

Table 1: Summary of datasets relevant to lyric–melody emotion analysis.

Dataset	**Lyrics**	**ABC / Symbolic**	**Metadata**	**Year**	**Primary Use**
Million Song Dataset	No	No	Yes	2011	Large-scale metadata and audio features for MIR research (HDF5 format)
musiXmatch Dataset	BoW only	No	Limited	2014	Lyrics-based text analysis aligned with MSD (TXT/CSV)
Last.fm Dataset	No	No	Yes	2002–09	Emotion, genre, and social tags at scale (CSV/JSON)
DEAM Dataset	Yes	No	Yes	2017	Continuous valence–arousal emotion labels (CSV/WAV)

Dataset	Lyrics	ABC / Symbolic	Metadata	Year	Primary Use
MoodyLyrics4Q	Yes	No	Partial	2015	4-class emotion classification benchmark (TXT/CSV)
Spotify Million Playlist	No	No	Yes	2018	Playlist modelling and recommendation (JSON)
Nottingham Dataset	No	Yes (ABC)	Minimal	∼1990s	Symbolic folk melody modelling (ABC/MIDI)
The Session	No	Yes (ABC)	Minimal	2000s	Irish traditional tune ABC corpus
Lakh MIDI Dataset	No	MIDI	Yes	2016	Large-scale symbolic music modelling, linked to MSD

No single dataset provides all three components—lyrics, ABC notation, and emotion labels—simultaneously. However, the Million Song Dataset serves as a hub through its interoperability with other resources: lyrics can be obtained via the musiXmatch Dataset using shared song identifiers, emotion and genre tags from Last.fm, and symbolic representations from the Lakh MIDI Dataset. The Nottingham Dataset and The Session provide substantial ABC notation corpora (the abcnotation.com archive alone contains approximately 800,000 tunes), though these focus on folk and traditional music and lack lyrics and emotion annotations. A viable strategy for constructing the dataset required for our research would involve linking multiple sources through the MSD's identifier system and supplementing ABC notation from dedicated archives.

We propose a data alignment pipeline as shown in Figure 1, which integrates three core datasets: the Million Song Dataset (MSD), Lakh MIDI Dataset, and MusiXmatch Dataset. At the heart of this pipeline is hdf5_getters.py, a utility provided by the MSD that enables structured access to rich metadata. This utility drives two parallel processing streams: one that filters and extracts song_id from the MSD to match with lyrics in the MusiXmatch Dataset, and another that uses track_id to retrieve corresponding MIDI files from the Lakh MIDI Dataset. The Filter and Convert modules (highlighted in green) are the custom components we developed to refine this process. The Filter module applies

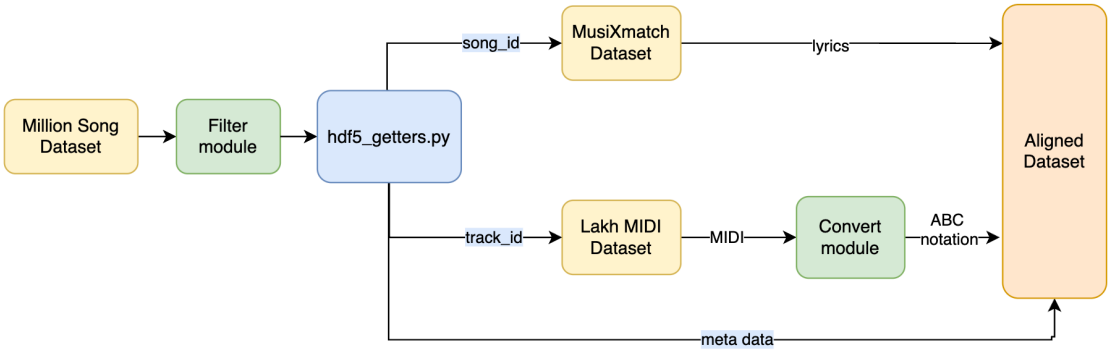

Figure 1: Data alignment

user-defined criteria—such as genre, artist, and style—to curate the MSD, while the Convert module translates MIDI files into ABC notation using Python libraries, making musical structure accessible for analysis.

While this pipeline effectively aligns lyrics, audio metadata, and symbolic music representations, it has a key limitation: it does not support temporal synchronization between lyrics and ABC notation. This alignment would require additional timestamp data, which is outside the scope of our current work, as our analysis focuses on separate, rather than temporally synchronized, examination of these modalities.

8 Discussion and Research Gaps

Several key findings emerge from the surveyed literature:

1. **Lyrics encode meaningful musical information.** Fell and Sporleder (2014) demonstrated that linguistic features of lyrics—including vocabulary diversity, structural repetition, and POS distributions—reliably predict genre, quality, and era. This establishes that lyric text is not merely decorative but carries systematic signals related to musical identity.

2. **Topic models capture latent lyrical semantics.** The LDA-based approach of Sasaki et al. (2014) provides a method for quantifying thematic content in lyrics beyond surface-level features, enabling similarity computation between songs on the basis of their underlying topics.

3. **Melody dominates but does not monopolise emotional expression.** Ali and Peynircioğlu (2006) found that melody tends to be the primary emotional carrier, with lyrics playing a modulatory role. However, this does not imply lyrics are emotionally inert—rather, the interaction between modalities is context-dependent.

4. **Combining modalities consistently improves emotion recognition.** Both Malheiro et al. (2013) and Rocha et al. (2013) found that fusing lyric and audio features outperforms either modality alone, suggesting that lyrics and melody provide complementary emotional information.

5. **Melodic contour features are particularly informative.** Rocha et al. (2013) showed that melodic audio features outperform standard spectral features, implying that pitch contour—the feature most naturally encoded in ABC notation—is a strong emotional indicator.

6. **ABC notation is a viable but underexploited modality.** Walshaw (2014) catalogued substantial ABC corpora and demonstrated pitch contour extraction from ABC, but no study to date has systematically compared ABC-derived melodic features with lyric-derived emotion features for the same songs.

The primary gap identified is the absence of research that directly aligns symbolic melodic representations (specifically ABC notation) with lyric-level emotion analysis within a unified framework. Existing MER studies use audio waveform features rather than symbolic notation, and existing ABC notation research focuses on duplication detection and corpus statistics rather than emotion. Our planned study aims to bridge this gap by constructing a paired dataset of lyrics and ABC representations, extracting emotion-relevant features from each modality, and quantifying the extent to which they elicit congruent emotional responses.

9 Conclusion

This review has surveyed the relevant literature across lyric-based text classification, topic modelling, symbolic music representation, and music emotion recognition. The evidence consistently indicates that both lyrics and melody carry emotional information, with melody generally serving as the dominant modality but lyrics providing meaningful complementary signals. Methodologically, SVM classifiers emerge as the most effective approach across studies, and feature engineering—particularly structural, stylistic, and melodic contour features—proves more informative than simple bag-of-words baselines.

The critical gap in the existing literature is the lack of studies that compare emotion features derived from ABC notation (a symbolic melodic representation) with emotion features derived from lyrics for the same songs. Our forthcoming research will address this gap by leveraging the interoperability of existing datasets to construct paired lyric–melody representations and by developing a framework for cross-modal emotion comparison at the song level.

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